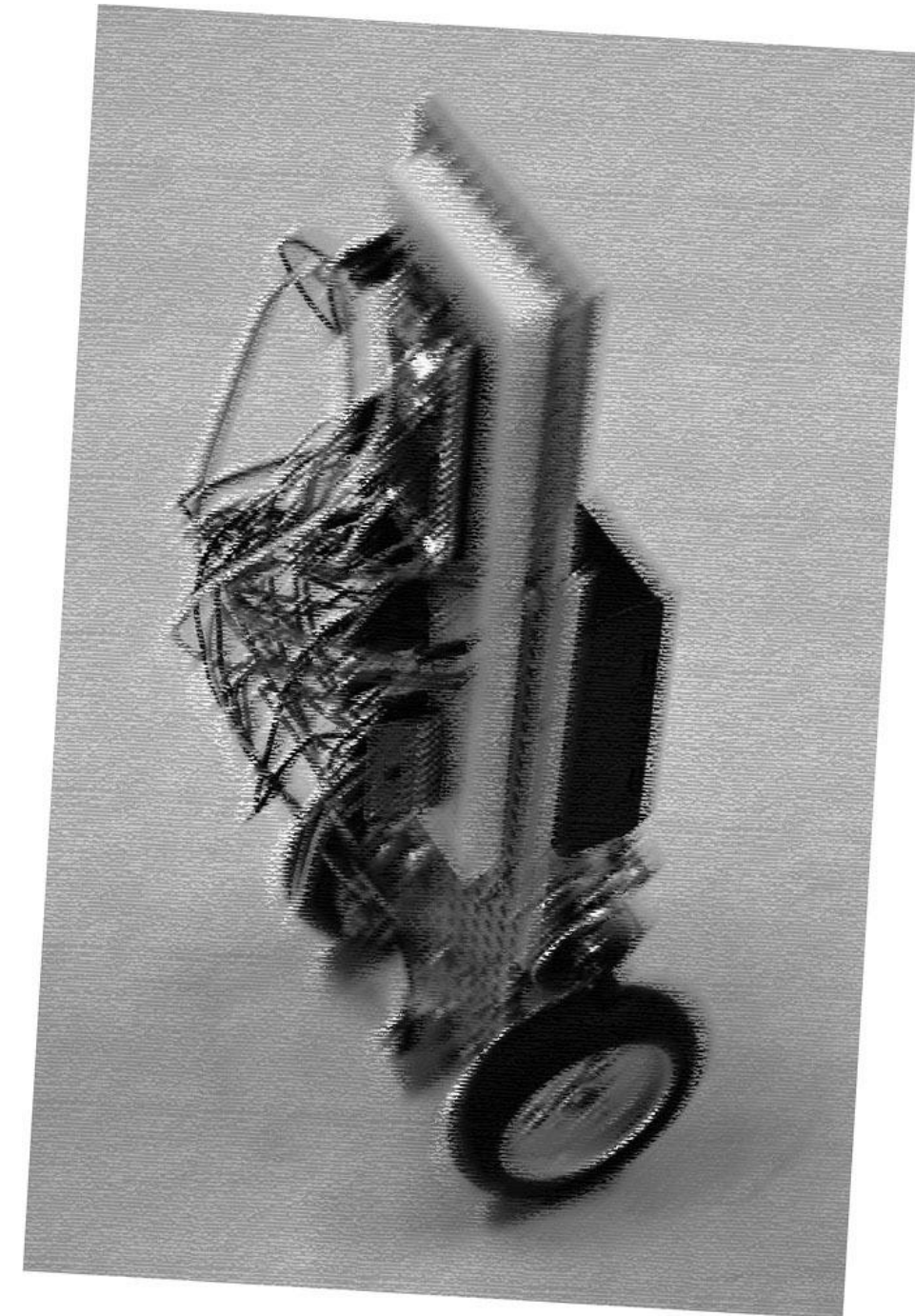


PRACTICAL EXERCISE · WEEK 3

Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

COURSE Department of Precision Engineering · 2026 S1S2
SESSIONS Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14



Timeline

1

Week 1 Standing

Assembly & bring-up
Angle-only **standing** demo

2

Week 2 Position control

Build a rotary encoder
Position control
Simulation in Simulink

3

Week 3 EXTENSION

Load & disturbance tests
Advanced control (LQR/MPC)
.....

4

Week 4 presentation

10-minute presentation
Live demo & Q&A
Final grade · 40%

Week 3 is for extended tasks, or you can continue working on the tasks of Week 1 and Week2

Week 3 - Extension Tasks

Reference Topics

1. Load Test

2. Enhanced Simulation

3. Advanced Controller

4. Slope Challenge

5. Back-and-Forth Motion

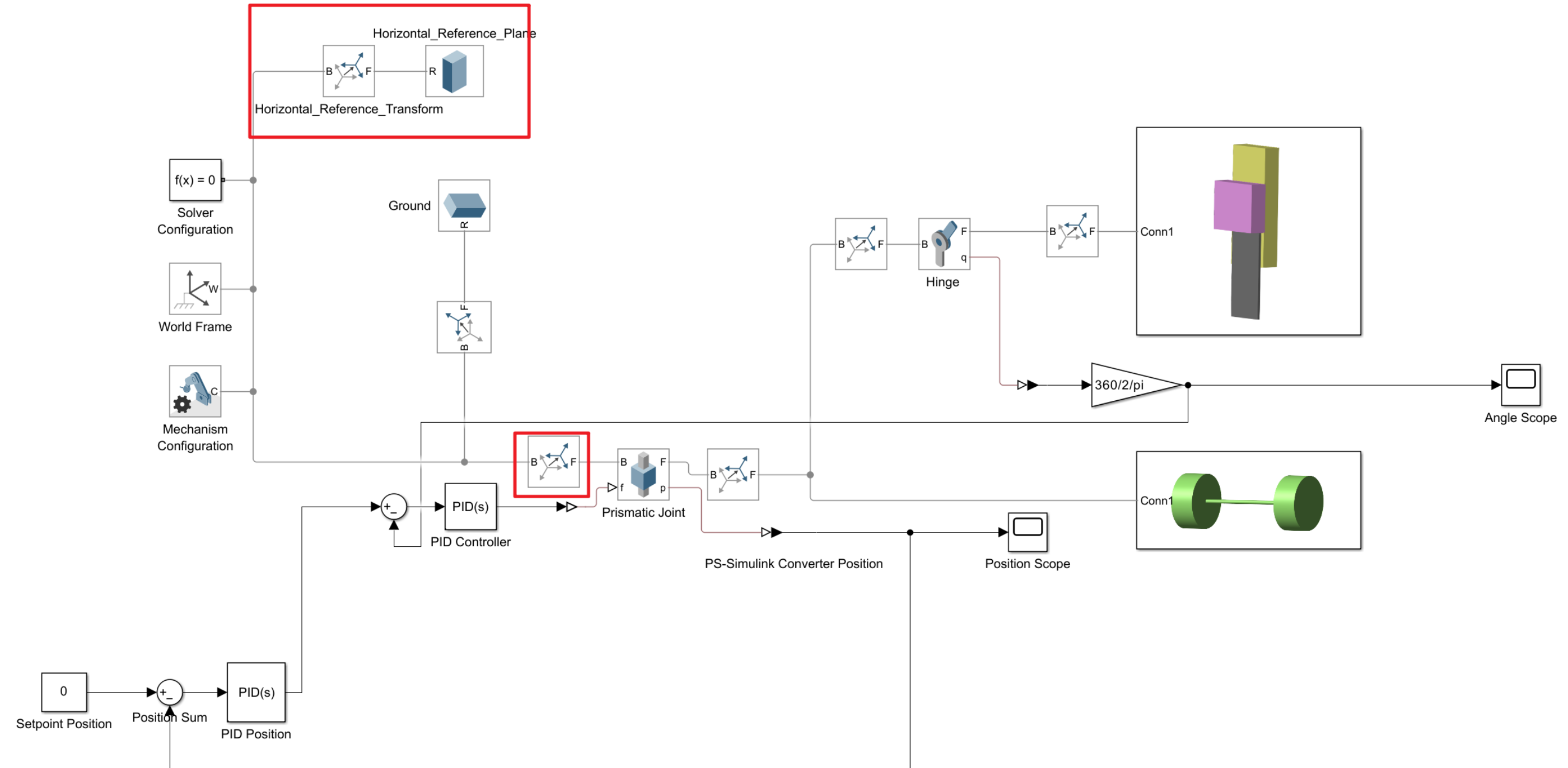
6. Maximum Disturbance

Assign the tasks among yourselves.

Can be done in hardware or simulation.

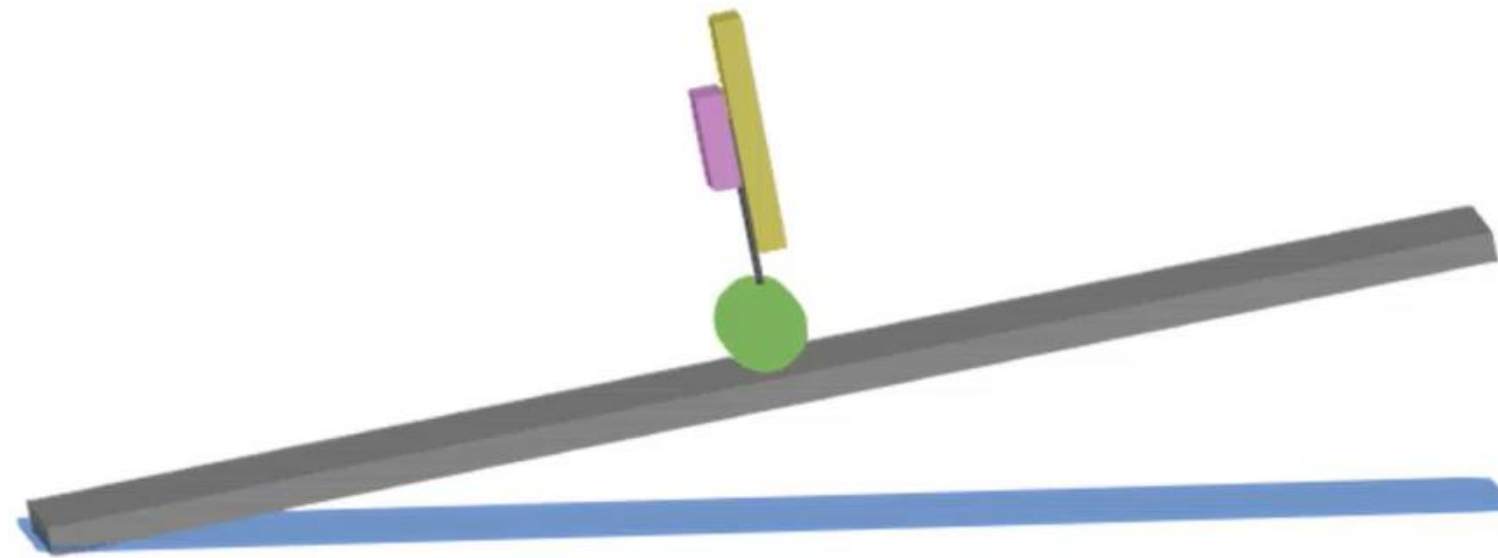
Week 3 Simulink Simulation - Slope Challenge

4



■ Week 3 Simulink Simulation - Slope Challenge

5



WEEK

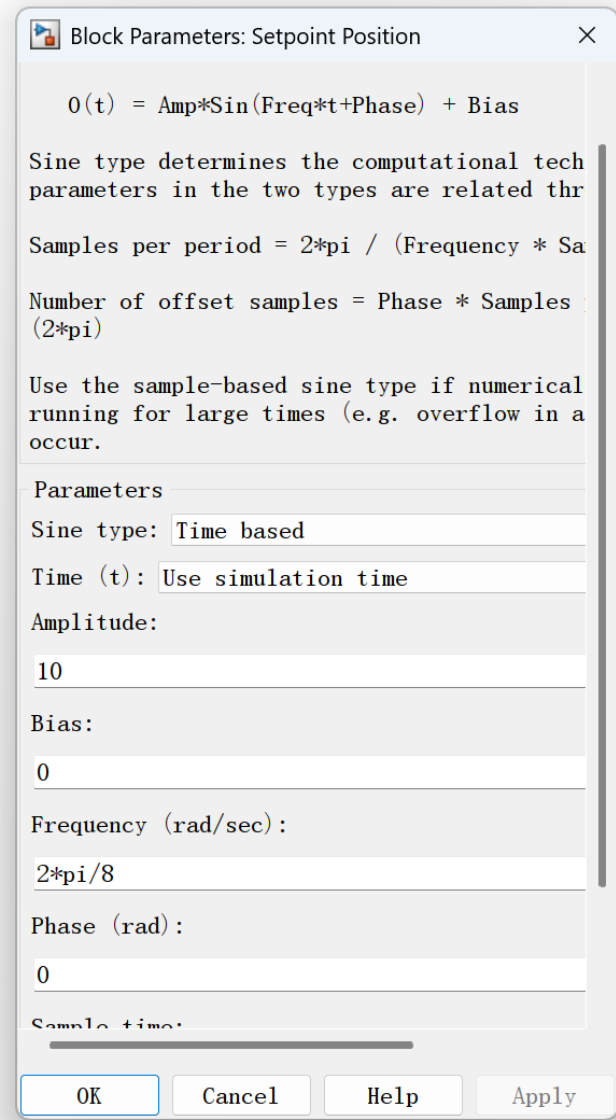
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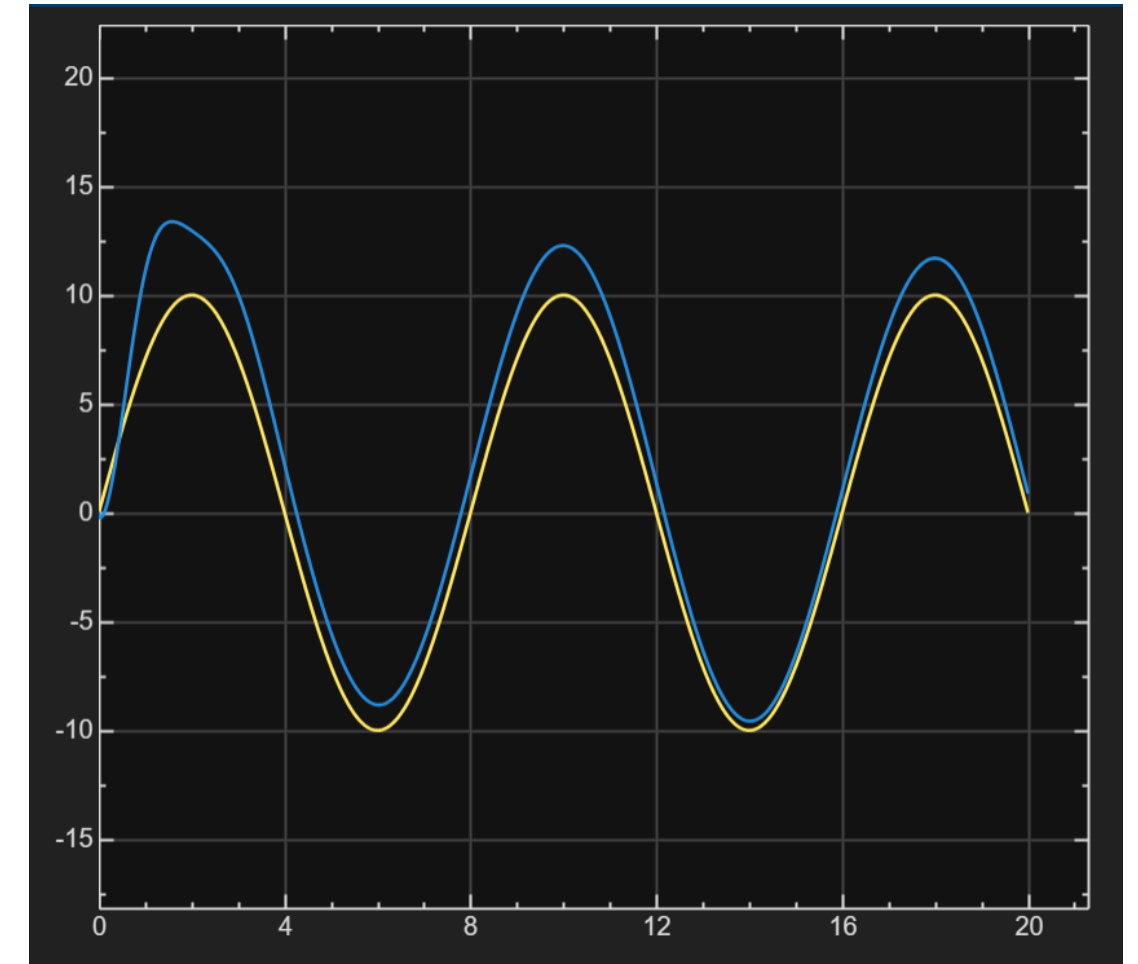
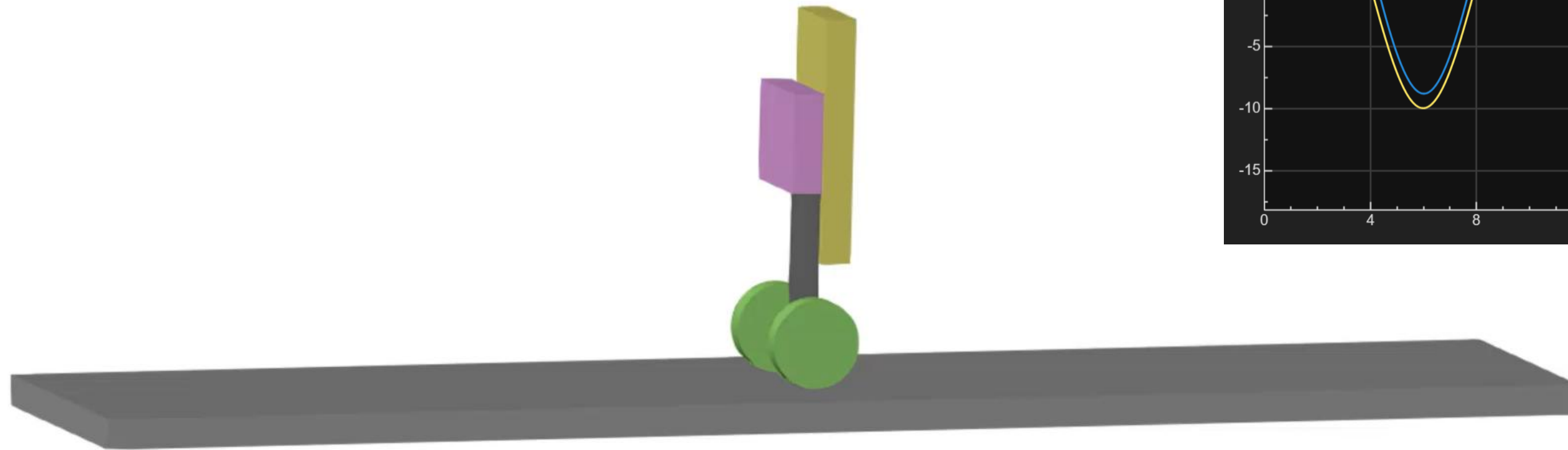


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■ Week 3 Simulink Simulation - Back-and-Forth Motion

8



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Week 3 Submission

Video 1:

The largest disturbance or load that your real wheeled inverted pendulum can withstand.

Video 2:

The largest disturbance that your simulation model can withstand under comparable conditions

One-page summary (PDF):

explain what maximum disturbance or load you applied, what you improved compared with the previous stage, and, if the system did not work, explain why it did not work.

DEADLINE

21:00

on the day of the class

Today — May 11

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Each group only needs one submission.

WEEK 4 · FINAL SHOWCASE PREVIEW

8 slides - 7 minutes + 3-minute demo + 5-min Q&A

01 Introduction Project goal, team task, robot configuration. State the problem and why it matters.	02 Method Modeling, control design, sensor usage, implementation strategy.	03 Experiment Test setup, conditions, evaluation procedure. How did you measure success?	04 Data Key plots, tables, numerical results — the evidence behind your verdict.	05 Extended Experiment Beyond the basic task — load test, parameter study, sim vs. real, alternative controller.
06 Results & Future Work What worked, what didn't, what to improve next. Honest assessment over hype.	07 What You Learned Reflect on lessons — technical, experimental, teamwork, debugging.	08 Teamwork & Roles Who did what — task split, collaboration, individual contributions and shared decisions.	+3 MINUTES Demo Video or Live Run A video or run the robot live	

WEEK 4 · FINAL SHOWCASE PREVIEW

8 slides - 7 minutes + 3-minute demo + 5-min Q&A

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Name	Date modified	
📁 fritzing	Apr 14	⋮
📁 matlab_simulink	Apr 14	⋮
📁 video	Apr 14	⋮
📁 appendix	Apr 14	⋮
📁 debug_tools	Apr 14	⋮
📁 .git	Apr 14	⋮
📄 Final_Presentation_Template.pptx 👤	10:59 AM	⋮
📄 Course-Introduction-Week-2.pdf 👤	Jun 8	⋮
📄 Course-Introduction-Week-1.pdf 👤	May 25	⋮
📄 Course-Introduction-Week-4.pdf 👤	May 18	⋮
📄 Course-Introduction-Week-3.pdf 👤	May 11	⋮

2026年6月15日
周一 11:03 (Local)

You can refer to the template we provided.